SIT292 Module 2 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	4/4	
Question 2	1/1	
Question 3	2/2	
Question 4	2/2	
Question 5	1/1	
Total	10/10	(100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 2 Self-Assessment
Session ID:	5031708919757769
Exam Start:	Sun Jul 20 2025 13:31:29
Exam Stop:	Sun Jul 20 2025 14:12:13
Time Spent:	0:40:44

Question 1

a)

Calculate the following (use the controls at the top to set the size of the matrix)

$$-4\begin{pmatrix} -3 & 4 & -3 \\ -5 & 4 & 5 \end{pmatrix} - 3\begin{pmatrix} -2 & -3 & -1 \\ 4 & 1 & 5 \end{pmatrix} =$$

Rows: 2 Columns: 3

$$\begin{pmatrix} 18 & -7 & 15 \\ 8 & -19 & -35 \end{pmatrix}$$ ✔

Expected answer: $\begin{pmatrix} 18 & -7 & 15 \\ 8 & -19 & -35 \end{pmatrix}$

.

Score: 2/2 ✔

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b)

Calculate the following

$$\begin{pmatrix} 4 & -2 \\ -3 & 1 \\ -4 & 1 \\ 2 & 3 \end{pmatrix}^T =$$

Rows: 2 Columns: 4

$$\begin{pmatrix} 4 & -3 & -4 & 2 \\ -2 & 1 & 1 & 3 \end{pmatrix}$$

Expected answer: $\begin{pmatrix} 4 & -3 & -4 & 2 \\ -2 & 1 & 1 & 3 \end{pmatrix}$

Score: 2/2

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Your answer is correct. You were awarded **2** marks.

You scored **2** marks for this part.

Question 2

Let $A = \begin{pmatrix} -3 & -1 & -1 & 0 & -3 \\ -1 & 1 & -2 & -3 & -1 \end{pmatrix}$

The matrix transformation of A is $T_A\colon \mathbb{R}^n \to \mathbb{R}^m$ for which numbers n and m?

$n =$ 5 Expected answer: 5 and $m =$ 2 Expected answer: 2

Score: 1/1 ✔

m

✔ Your answer is correct. You were awarded **0.5** marks.

n

✔ Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 3

Let $A = \begin{pmatrix} -2 & 1 & 0 \\ -2 & 2 & 0 \\ 0 & -3 & 4 \end{pmatrix}$

For a general vector $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, determine $T_A(x)$. Input x_1 as `x_1` or `x1`, etc. Use the same notation for all variables, don't mix it.

$T_A(x) = ($

-2x1+x2 $-2x_1 + x_2$ ✔

Expected answer: -2x_1 + x_2 $-2x_1 + x_2$

-2x1 + 2x2 $-2x_1 + 2x_2$ ✔

Expected answer: -2x_1 + 2x_2 $-2x_1 + 2x_2$

-3x2 + 4x3 $-3x_2 + 4x_3$ ✔

Expected answer: -3x_2 + 4x_3 $-3x_2 + 4x_3$

$)$

Score: 2/2 ✔

✔ Your answer is correct. You were awarded **2** marks.

You scored **2** marks for this part.

Question 4

Let $A = \begin{pmatrix} 2 & -2 & 2 & 3 \\ 3 & -1 & 5 & -4 \end{pmatrix}$, $B = \begin{pmatrix} 4 & -3 & 1 \\ 5 & -4 & 2 \\ -4 & -4 & -2 \\ 2 & 0 & 1 \end{pmatrix}$.

Compute $AB =$

Rows: 2 Columns: 3

$$\begin{pmatrix} -4 & -6 & -3 \\ -21 & -25 & -13 \end{pmatrix}$$ ✔

Expected answer: $\begin{pmatrix} -4 & -6 & -3 \\ -21 & -25 & -13 \end{pmatrix}$

Score: 2/2 ✔

✔ Your answer is correct. You were awarded **2** marks.

You scored **2** marks for this part.

Question 5

Consider the following nutrition table.

	Fat	Sodium	Carbohydrates
Banana	0.2 g	1.2mg	27 g
Cornflakes (bowl)	0.3g	236 mg	24g
Milk (cup)	3 g	110.9mg	1.2 g

The information in the table is represented as a matrix A, i.e.,

$$\mathrm{A} = \begin{pmatrix} 0.2 & 1.2 & 27 \\ 0.3 & 236 & 24 \\ 3 & 110.9 & 16 \end{pmatrix}$$

Suppose we have a points system that scores 0.9 points per gram of fat, 0.8 per mg of sodium, and 2 per gram of carbohydrates. Which of the following will give the total points scored by a banana, bowl of cornflakes, and cup of milk?

- ○ $\begin{pmatrix} 0.9 & 0.8 & 2 \end{pmatrix} A$
- ○ $\begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix} A$
- ◉ $A \begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix}$
- ○ $A \begin{pmatrix} 0.9 & 0.8 & 2 \end{pmatrix}$

✔

Expected answer:

- ○ $\begin{pmatrix} 0.9 & 0.8 & 2 \end{pmatrix} A$
- ○ $\begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix} A$
- ◉ $A \begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix}$
- ○ $A \begin{pmatrix} 0.9 & 0.8 & 2 \end{pmatrix}$

Score: 1/1 ✔

✔ You chose a correct answer. You were awarded **1** mark.

You scored **1** mark for this part.
